

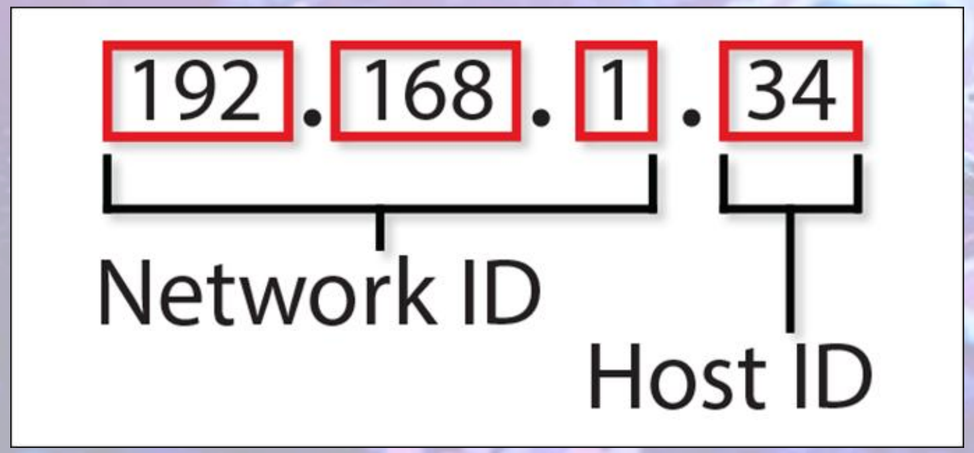
Network Basics
Security Essentials

Learn More

What is an **IP Address ?**



Internet Protocol Address is feature that manages and **prioritizes network traffic** to ensure that critical applications receive the necessary bandwidth and latency requirements, optimizing performance for specific data types and reducing congestion.



An IP Address is like the **home address of a device on a network.**
It helps identify where data should go.

Example: 192.168.1.34
When you send or receive data, it uses IP addresses to find the correct device.

Types of IP Address **Based on IP Version**



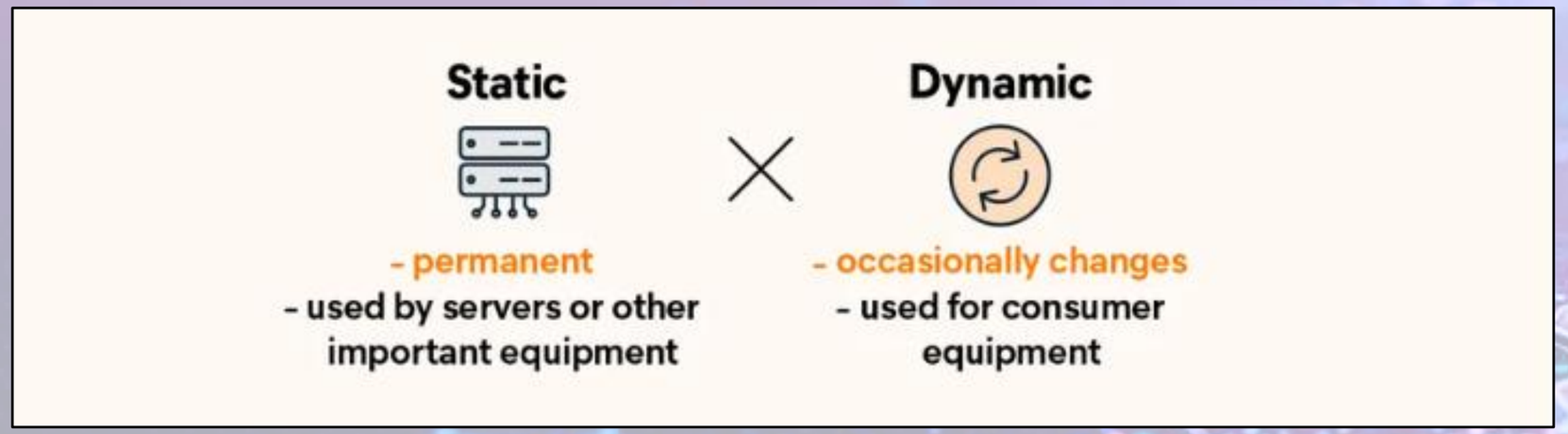
IPv4 (Internet Protocol Version 4)

- The most common version, using a 32-bit address format, represented by four numbers (0-255) separated by dots (e.g., 192.168.1.1).
- Supports approximately 4.3 billion unique addresses.

IPv6 (Internet Protocol Version 6)

- A newer version designed to address the limitations of IPv4, using a 128-bit address format with eight groups of four hexadecimal digits separated by colons (e.g., 2001:0db8:85a3:0000:0000:8a2e:0370:7334).
- Can support an immense number of unique addresses, making it future-proof as more devices connect to the Internet.

Types of IP Address **Based on Assignment Type**



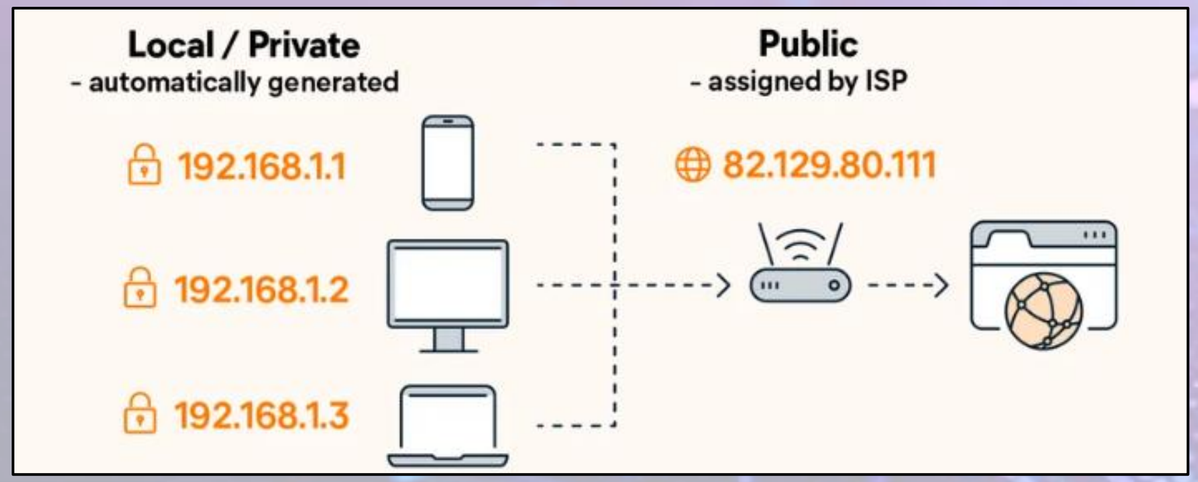
Static IP Address

- A fixed IP address that does not change and is manually assigned by a network administrator.
- Commonly used for servers, printers, and other devices needing a constant address.

Dynamic IP Address

- An IP address assigned by a DHCP (Dynamic Host Configuration Protocol) server and can change over time.
- Typically used for most personal devices, like smartphones, laptops, and home computers, to optimize address allocation.

Types of IP Address **Based on Network Scope**



Public IP Address

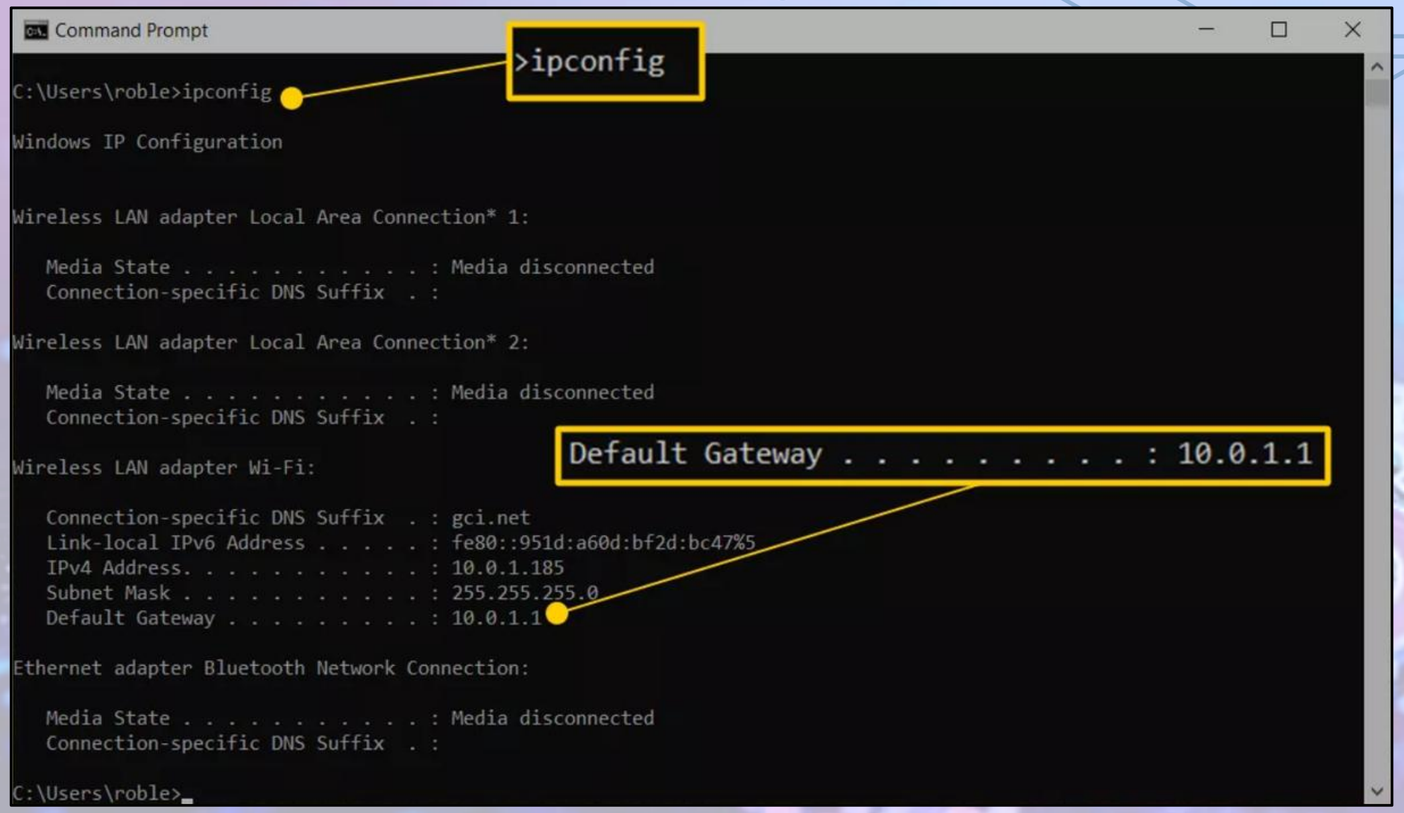
- A unique IP address assigned to a device that connects directly to the Internet.
- Visible to external networks and allows communication over the Internet (e.g., a web server's IP address).

Private IP Address

- Used within private networks (e.g., a home or office) and not routable on the public Internet.
- Reserved ranges for private IPs include:
  - 10.0.0.0 – 10.255.255.255
  - 172.16.0.0 – 172.31.255.255
  - 192.168.0.0 – 192.168.255.255

Types of IP Address

Based on Functionality



Default Gateway

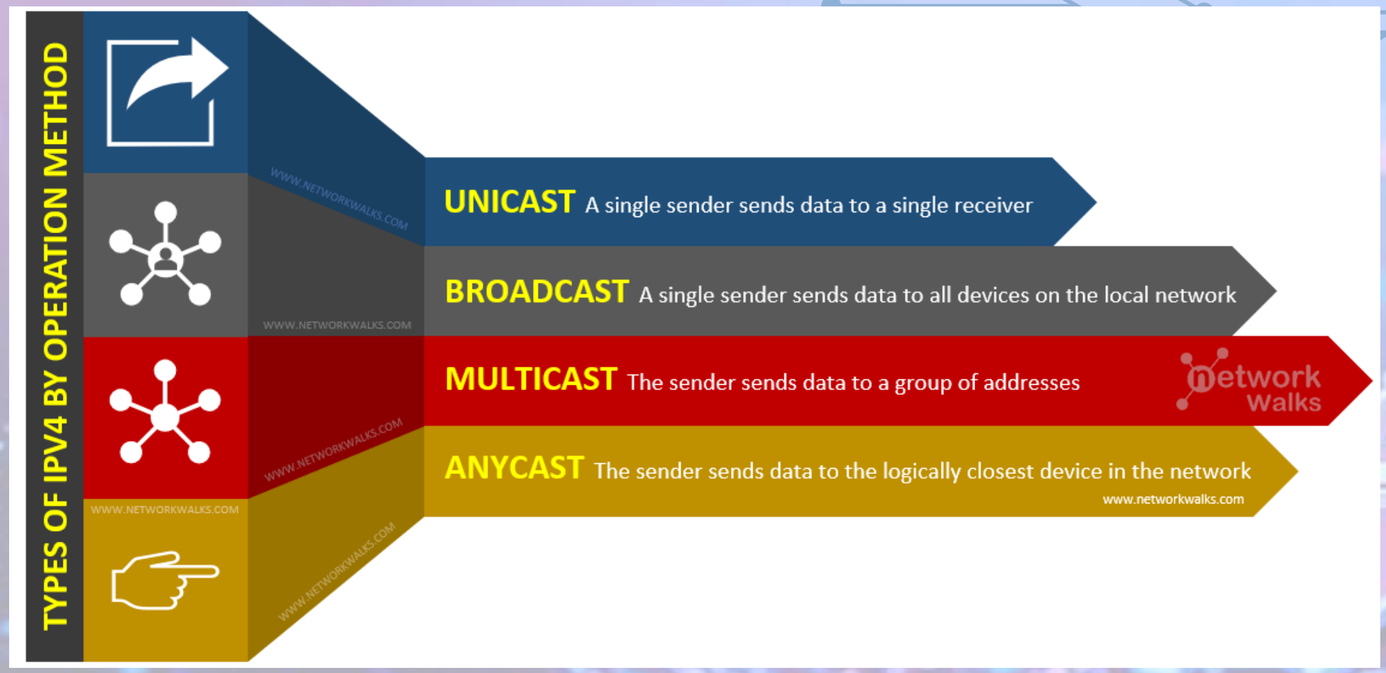
The IP address of a router in a network that serves as an access point to other networks, especially for connecting to the Internet.

Loopback IP Address

- A special IP address (127.0.0.1 in IPv4) that points back to the device itself, often used for testing network applications locally.

Types of IP Address

Based on Communication Type



Unicast IP Address

Used for one-to-one communication, where data is sent from one device to a specific device.

Multicast IP Address

Used for one-to-many communication, where data is sent from one device to a group of devices (used in applications like streaming).

Broadcast IP Address

Used for one-to-all communication, where data is sent from one device to all devices on the same network.

Different classes of **IP addresses**

Class	IP Address Range	Subnet Mask	Number of Networks	Hosts per Network	Typical Use
A	1.0.0.0 to 126.255.255.255	255.0.0.0	128	16,777,214	Large organizations, ISPs
B	128.0.0.0 to 191.255.255.255	255.255.0.0	16,384	65,534	Medium to large organizations
C	192.0.0.0 to 223.255.255.255	255.255.255.0	2,097,152	254	Small organizations, LANs
D	224.0.0.0 to 239.255.255.255	N/A	N/A	N/A	Multicast groups
E	240.0.0.0 to 255.255.255.255	N/A	N/A	N/A	Experimental or research purposes

- **Class A** addresses are typically used by large networks such as multinational corporations or governments, providing a vast number of hosts.
- **Class B** is suited for medium-sized networks, allowing for a reasonable number of hosts.
- **Class C** is common in small networks, such as local area networks (LANs).
- **Class D** is reserved for multicast groups, allowing a single sender to transmit data to multiple receivers.
- **Class E** is reserved for experimental use and is not commonly implemented in regular networking.
- Classes A, B, and C are the main classes used for public and private IP addressing, while D and E serve specific purposes in network communications.

Identify the type of IP address shown below:

192.168.1.17

I v

ANSWER!

192.168.1.17

I P v 4

Identify the type of IP address shown below:

127.0.0.1

P A C K

ANSWER

127.0.0.1

L O O P B A C K

Identify the type of IP address shown below:

FIXED IP

	T	A				C

ANSWER

FIXED IP

S T A T I C

Identify the type of IP address shown below:

2405:8d40:405c:8f6:dc3f:b77a:3135:a2ac

IPv6

ANSWER

2405:8d40:405c:8f6:dc3f:b77a:3135:a2ac

I		V	

Identify the type of IP address shown below:

93.230.91.222

B L

ANSWER

93.230.91.222

P U B L I C

Identify the type of IP address shown below:

RANDOM IP

	Y		A	M			

ANSWER

RANDOM IP

D Y N A M I C

Identify the type of IP address shown below:

192.1.2.3

C L A S S

ANSWER

192.1.2.3

CLASS C

Identify the type of IP address shown below:

1.6.5.4

C L A S S

ANSWER

1.6.5.4

CLASS A

Identify the type of IP address shown below:

128.8.7.9

C L A S S

ANSWER

128.8.7.9

CLASS B

DNS
Domain Name System

DNS is like a **phonebook of the internet.**
It **translates** website names into IP addresses.

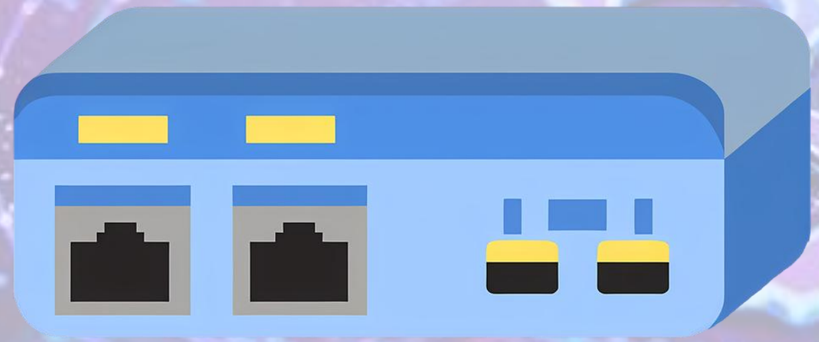
Example:
You type google.com → DNS finds its IP address → your computer connects to it.

Without DNS, **you would need to memorize IP addresses of websites.**



Ports

Ports are like **doors or channels** on a device.
They tell the system what type of service the data is for.



Common ports:

- 80 → HTTP (web browsing, not secure)
- 443 → HTTPS (secure web browsing)

So even if data arrives at the right IP, the port decides which application will handle it.

Common Targeted **Vulnerable Ports**



21 – FTP (File Transfer Protocol)

- Used for file transfer
- Vulnerability: Sends data (including passwords) in plain text
- Attack: Credential sniffing, brute force

22 – SSH (Secure Shell)

- Used for remote login (secure)
- Vulnerability: Weak passwords
- Attack: Brute force login attempts

23 – Telnet

- Used for remote access (old method)
- Vulnerability: No encryption
- Attack: Easy credential theft

25 – SMTP (Email Sending)

- Used to send emails
- Vulnerability: Open relay misconfiguration
- Attack: Spam, phishing campaigns

53 – DNS

- Used for domain name resolution
- Vulnerability: Can be abused
- Attack: DNS amplification (DDoS), spoofing

110 – POP3 (Email Receiving)

- Used to receive emails
- Vulnerability: Often unencrypted
- Attack: Credential sniffing

139 / 445 – SMB (Windows File Sharing)

- Used for file and printer sharing
- Vulnerability: Common target in Windows systems
- Attack: Ransomware (e.g., WannaCry), lateral movement

3389 – RDP (Remote Desktop Protocol)

- Used for remote desktop access
- Vulnerability: Exposed to internet
- Attack: Brute force, ransomware entry point

3306 – MySQL

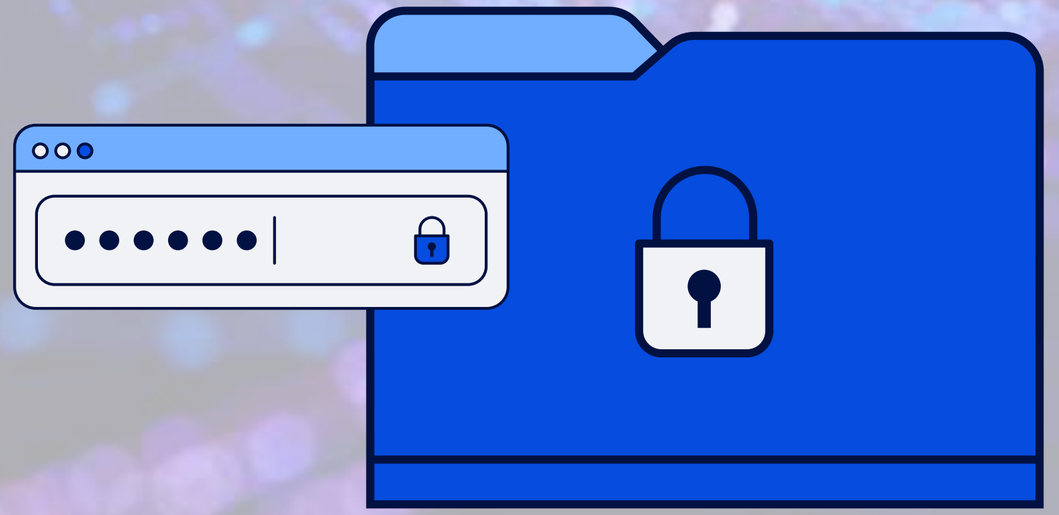
- Used for database access
- Vulnerability: Weak credentials, exposed DB
- Attack: Data theft

8080 – Alternative HTTP

- Used for web apps or proxies
- Vulnerability: Often less secured than port 80
- Attack: Web exploits

Redis (Port 6379)

- Attacker scans internet for open port 6379
- Finds Redis with no password
- Connects and injects malicious commands
- Gains control or installs malware



Identify the
hidden word base on the image shown:



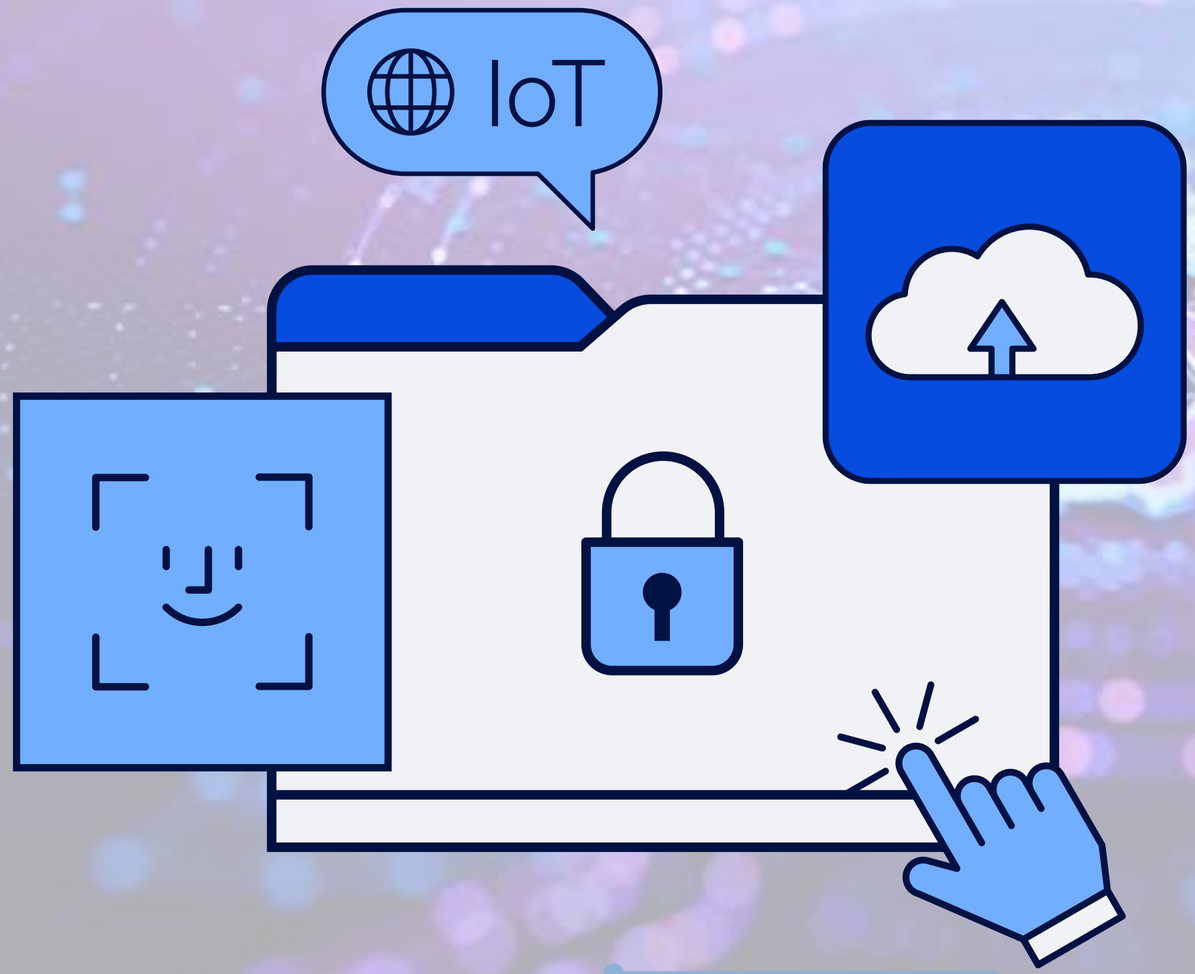
HANDSHAKE

A handshake in networking is the process of two devices agreeing on how to communicate before sending data.

Think of it like:

“Hello, can we talk?”

“Yes, here’s how we will talk.”



PREPARED BY: ADONIS OLIVER C. MACEDONIO

TCP Handshake

This is called the Three-Way Handshake in TCP.

Step-by-step:

- SYN (Synchronize)
  - Client → Server
  - “I want to connect”
- SYN-ACK (Synchronize + Acknowledge)
  - Server → Client
  - “Okay, I’m ready”
- ACK (Acknowledge)
  - Client → Server
  - “Connection established”

After this, data can now be sent safely.

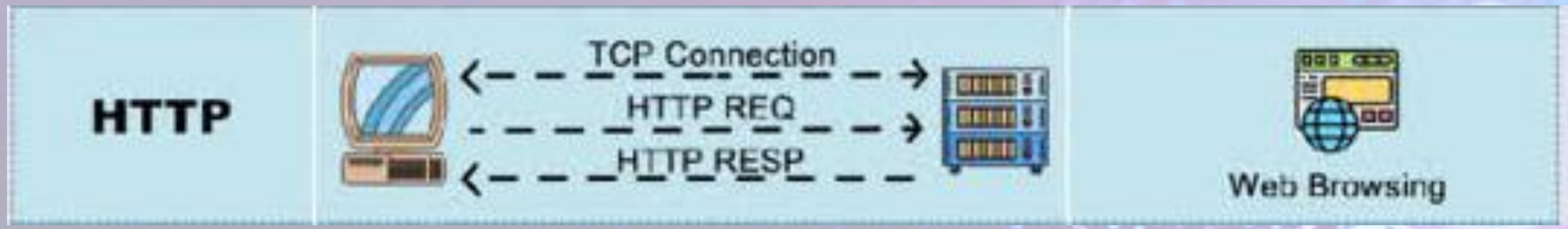


Understanding **Internet Protocol**

The internet is an indispensable part of our daily lives. From browsing websites to sending emails and streaming videos, countless activities rely on a complex web of rules and procedures known as internet protocols. These protocols are the unsung heroes that ensure data travels efficiently and securely across networks.



HTTP (Hypertext Transfer Protocol)

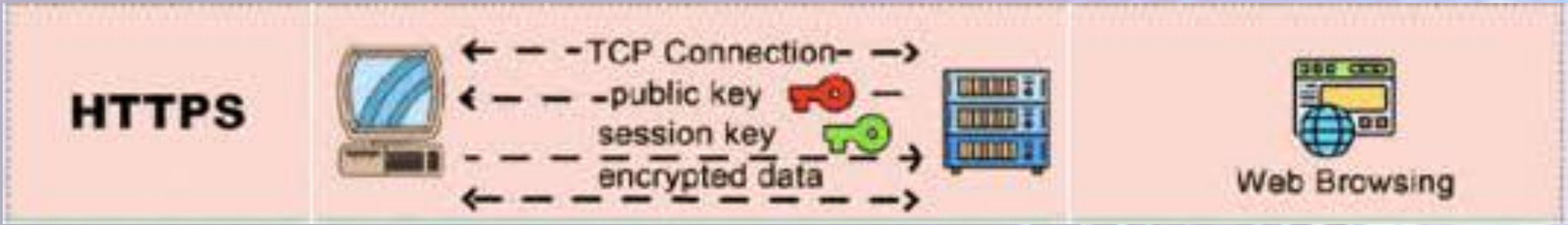


- Used for loading websites
- Data is sent in plain text (not secure)

Example: http://example.com

Can be intercepted by attackers

HTTPS (Hypertext Transfer Protocol Secure)

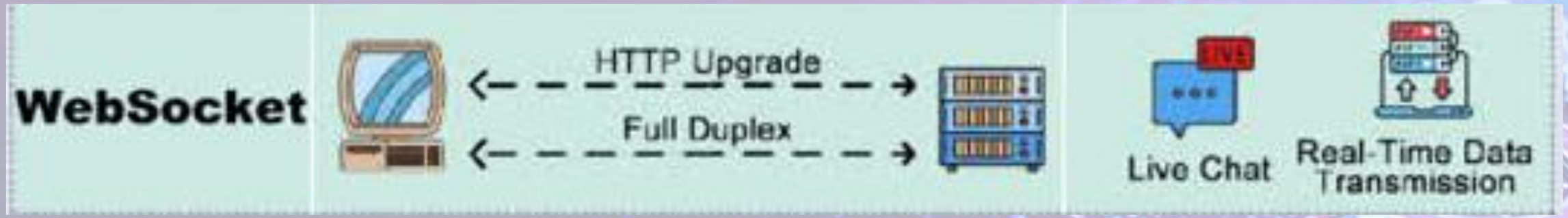


- Secure version of HTTP
- Uses encryption (SSL/TLS)

Example: https://example.com

Protects passwords, logins, and sensitive data

WebSocket

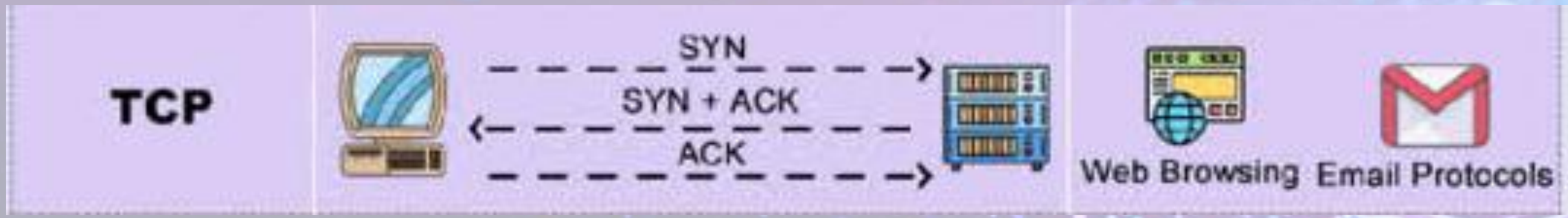


- Allows real-time, two-way communication
- Connection stays open

Used in:

- Chat apps
- Live notifications
- Online games

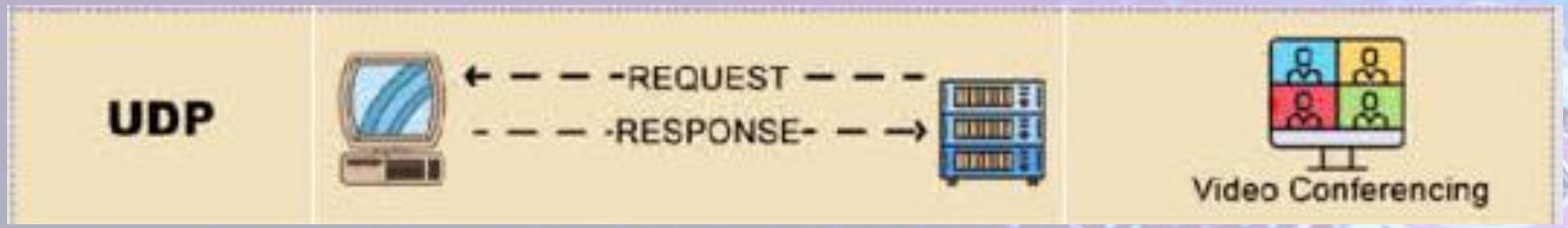
TCP (Transmission Control Protocol)



- Reliable communication
- Makes sure data is complete and correct
- Slower but accurate

Used in: web browsing, email, file transfer

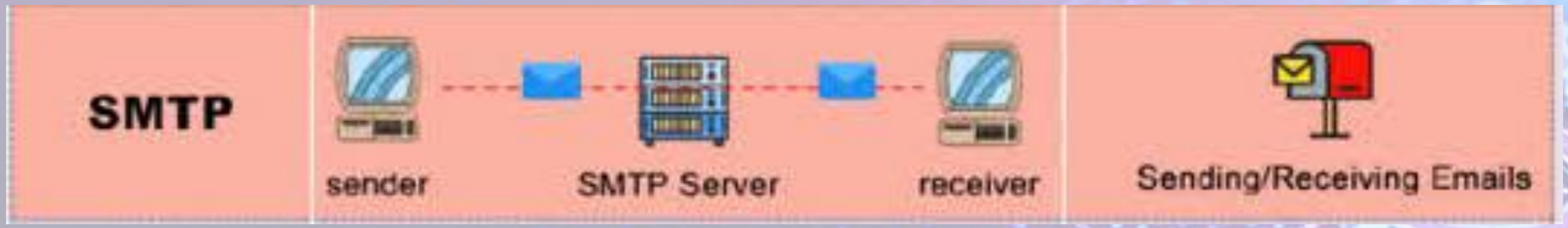
UDP (User Datagram Protocol)



- Fast but not reliable
- No checking if data is lost

Used in: video streaming, online games

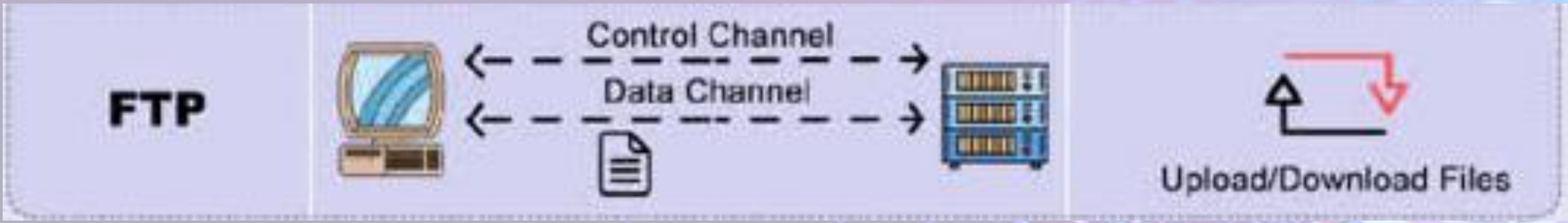
SMTP (Simple Mail Transfer Protocol)



- Used to send emails

Example: when you click “Send” in email

FTP (File Transfer Protocol)



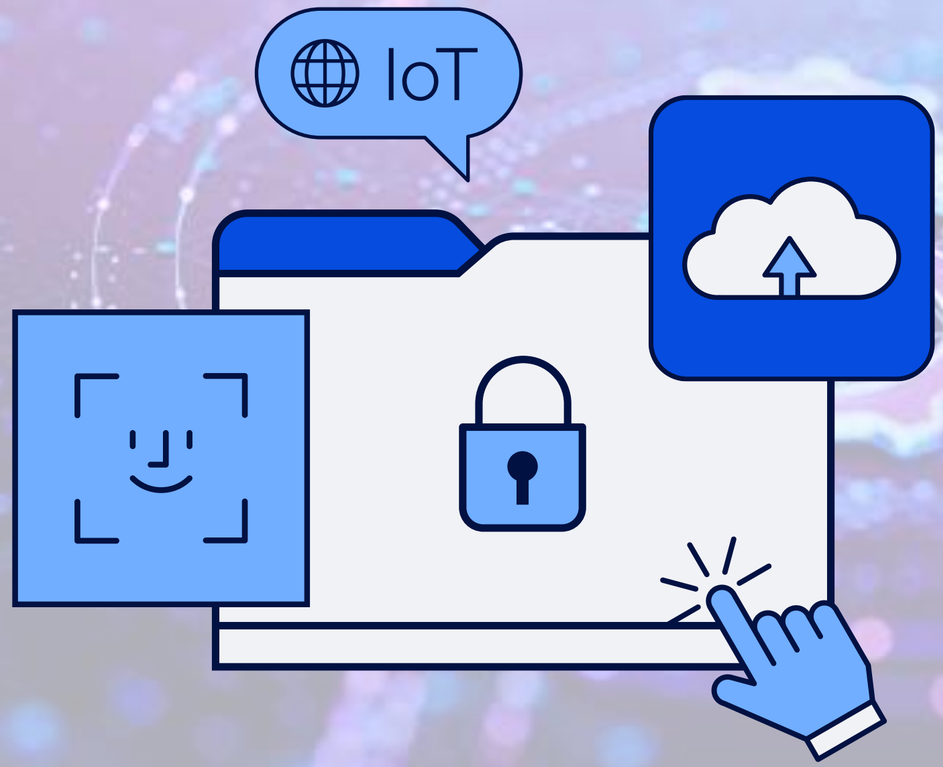
- Used to transfer files between computers

Example: uploading files to a website server

Not secure unless configured with encryption

Activity # 6
CTM Simulation (Understanding Ports)

This activity is a **Cyber Threat Monitoring Simulation** designed for you to practice identifying and mitigating cyber threats.



You will run a Python program that:

- Displays **10 random realistic cyber attacks**
- Shows **logs for each attack**
- Requires students to select the **correct port to mitigate each threat**
- Measures **time taken, score, and rating**
- Generates a **log report file** after completion

1. Clone or Download the Repository

1. Go to the repository: `CTM_Activity 6 VM`
    - https://github.com/xen0byt3/CTM-Activity-6
2. Click **Code → Download ZIP** or clone using Git:
    - git clone <repository_url>

Extract the ZIP (if downloaded) or navigate to the folder (if cloned).

2. Run the Simulation

1. Make sure Python 3 is installed on your machine.
2. Open a terminal or command prompt in the repository folder.
3. Run the simulation:
    - python main.py
4. Follow the on-screen instructions:
5. Enter your Name and Cyber Handle
6. Review the logs and threat descriptions
7. Select the correct port for each threat (1–4)
8. Each question has a time limit of 10 seconds

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5. Pull Request Instructions (GitHub)

1. Create a new branch in your local repo (e.g., student_name_activity6):
    - git checkout -b student_name_activity6
2. Copy or move your cyber_log_report.txt file into the repo folder.
3. Stage and commit your changes:
    - git add cyber_log_report.txt
    - git commit -m "Add log report for CTM Activity 6"
4. Push the branch to GitHub:
    - git push origin student_name_activity6
5. Go to the repository on GitHub and create a Pull Request for your branch.

Thank You for Listening!

